Jialin Li

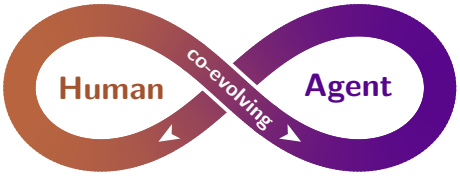
jialin.li@nyu.edu | Homepage | Google Scholar

Education

New York University Sept. 2024 – May 2029 (Expected)
Ph.D. in Computer Science, NYU Abu Dhabi Global Ph.D. Fellow *New York, NY, USA*
Advisor: Hanan Salam

New York University Shanghai Sept. 2019 – Jan. 2023
B.S. in Data Science *Shanghai, China*

Research Interests



Research Focus: People-Aware LLM Agents. I build and benchmark personal agents that adapt to individual personality, context-dependent interaction preferences, and continuous user feedback.

- User multimodal affect modeling (MOCSE, PerAug, PAC Survey)
- Tracking user preference states through preference adaptation and long-horizon memory (PrefIx, MemPABench)
- Building adaptive harnesses by turning users' interactions into updates for alignment & harness (Companion Agents @ Tencent)

Experience

Tencent Hunyuan, Foundation Model Team May 2026 – Sept. 2026
Algorithm Research Intern, In-Game Companion Agents *China*

- **Interactive Data Synthesis:** Context drift and unnatural TTS cadence in live gameplay cannot be resolved by existing training dialogues due to the lack of in-game state grounding. Built an interactive synthesis pipeline involving in-game states. It distills diverse player personas at realistic distributions and uses affective-cognitive profiles to enforce NPC's role consistency. Generated 10K+ multi-turn dialogues.
- **Synthesis Evaluation & Data Mixing:** Maintained an automated evaluation pipeline with a **two-layer rubric**, where gating metrics (safety, logic) set the quality floor and uplift metrics (spoken style, plot progression) raise the UX ceiling. After calibrating rubric anchors against human ratings, scaled evaluation with an automated judge to guide data inclusion and mixing and select targeted behaviors for RL reward modeling.
- **Real-Time State & Layered Memory:** In PUBG, rapid game state changes often led to stale context and thus causing model hallucinations. ❶ Consolidated game state, in-match events, dialogue history, and bot's action outcomes into a shared, structured context that updates incrementally, cutting mismatched triggers by 8%. ❷ Designed a **five layered memory** framework for more efficient summarization. The four levels are: `L0` real-time state, `L1` recent events, `L2` single-match context, `L3` cross-match history, and `L4` game knowledge and skills. Further constructed training samples on accurate memory recall and dialogue continuation.
- **Implicit-Feedback Reward & Multi-Turn RL:** Performed SFT to inject the bots persona settings and world knowledge. Designed a **multi-objective reward model** calibrated on player behavioral signals (mic open frequency, chat continuity) and trained the bot with **offline PPO and online GRPO** toward multi-turn long-horizon goals.
- **Attribution & Data Flywheel:** Built a live bad-case analysis pipeline processing 20K sessions a day. Analyzed root cause by system layer. Closed a feedback → attribution → retraining **data flywheel** for targeted data iteration, supporting a product serving 8M total users with a peak DAU of 500K.

KPMG 2021
AI Solution Engineer Intern *Shanghai, China*

- Supported **conversational AI** development for automotive sales; collaborated with clients to define metrics across 10 features and maintained systematic error tracking.
- Scraped 11M+ dialog snippets from TikTok and other platforms; annotated for conversation flow, topics, and sentiment to create **datasets for LLM fine-tuning**.
- Implemented **guardrails** including length limits, stop word filtering, and confidence calibration; contributed to knowledge graph construction for vehicle attributes.

Publications

ACL Findings 2026 **PrefIx: Understand and Adapt to User Preference in Human-Agent Interaction**
J. Li, Z. Chen, H. Luo, H. Salam. Code

- **Problem:** LLM agents nowadays can solve tasks correctly, but their interaction still frustrates users in various ways, e.g. excessive confirmations. Existing benchmarks only evaluate task accuracy, missing the flaws in interaction.
- **Approach:** Introduced ***interaction preferences*** and **Interaction-as-a-Tool (IaaT)**. IaaT frames models interaction behaviors (confirm, disambiguate, escalate etc.) as structured tool calls, to make them evaluable with standard tool-use benchmarks. We test this across 31 preference in four dimensions (transparency, interaction pace, strategic proactiveness, robustness) with simulated users. Evaluated on seven UX dimensions by a panel of four LLM judges.
- **Findings:** Our agents match baseline accuracy while improving UX by 7.6% and alignment by 18.5%. Among the four preference dimensions, transparency was the easiest to satisfy, while robustness was the hardest.

Under Review **Why Bother Me Again? Benchmarking Personal Agent Memory for Interaction Preferences**
J. Li, Z. Chen, H. Luo, H. Salam. Code Webpage

- **Problem:** In long-term assistance, personal agents should track users' changing interaction (IX) preferences across contexts and over time to keep the interactions satisfying. However, current memory benchmarks have not been evaluating evolving IX preferences in long-horizon usage.
- **Approach:** Built an interactive environment for **on-policy memory evaluation** MemPABench. Presented four grounded simulated users with 28 interaction preferences across work and personal contexts. They used the testing agents over 99 sessions across four months, including five preference shifts. Evaluated four memory systems across various LLM backbones on preference selection, delivered behavior, and user-perceived interaction quality.
- **Findings:** Retrieval-based memory (Mem0, MemOS, RAG) merely matches the no-memory baseline. File memory performs better, but it overgeneralizes in misapplying preferences in the wrong contexts. Worse, agents struggle with execution. Even given the right preference, they deliver the intended behavior only 59% of the time.

ACII 2025 (Oral) **Improving Personalisation in Valence and Arousal Prediction using Data Augmentation**
J. Li, M. Nwadike, H. Salam.

- **Problem:** Per-person valence and arousal models have little data for each individual.
- **Approach:** Proposed a data-driven **sample-level augmentation** that pulls similar segments from multi-person data to adapt the target model. Improved CCC on personalization benchmark without changing model architectures.

ACM MM 2025 Workshop **Personality-Aware Engagement Prediction in Online Learning**
J. Li, G. Sharma, H. Salam.

- **Problem:** Vision-based engagement prediction ignores individual differences, and existing datasets are mostly lab-collected, one-dimensional, and lack self-reports and personality.
- **Approach:** Built MOCSE, an in-the-wild dataset of Zoom lectures (72 students, 36.5 hours) with self-reported and observer engagement labels plus Big Five personality; conditioning VLMs on personality improves prediction.

arXiv 2023 **Beyond One-Size-Fits-All: A Survey of Personalized Affective Computing in Human-Agent Interaction**
J. Li, M. Elgarf, A. Waleed, H. Salam.

- First survey of personalized affective computing, with a taxonomy of data- and model-level approaches.

Skills

Evaluation & Data: Interactive benchmark and environment design, LLM-as-a-judge, multi-turn data synthesis
Post-Training: SFT, reward modeling, PPO, GRPO
Programming: Python, Rust, Shell, SQL; PyTorch, Transformers; Docker, Git, Linux, WandB

Honors & Awards

NYU Abu Dhabi Global Ph.D. Fellowship, New York University — 2024 – Present
University Honors Scholar, New York University — 2023
Dean's List of Academic Year, New York University — 2021 – 2022
Dean's Undergraduate Research Fund, New York University — 2020 – 2021

Service

Reviewer: ACM MM 2025, SuSS 2025, ACL 2026, ICLR 2026, IEEE TAC, EMNLP 2026
Outreach: Association for the Advancement of Affective Computing (AAAC) DEI Committee, 2025